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# Efficient and secure e-voting scheme using elliptic curve cryptography

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## Abstract

Voting allows the people to elect their representative and express their preferences. In modern day democracy, voting is one fundamental and most important tool for election. To strengthen this process, efforts must be made to achieve a confirmable and transparent voting system. Naturally, the veracity of the election process is of ultimate importance for the honor of the democracy itself. The online voting or e-voting system is a cost-effective procedure, which saves a lot of money and time spent for organizing the election. However, security, integrity of data and privacy of the voter are the increasing concerns in this context. In this paper, we have done a thorough literature survey of the various latest schemes in this context and found that the schemes prone to several security and privacy threats. Moreover, a major issue with all these schemes is that they are inefficient in terms of computation and communication overheads. With this motivation, we present a novel and efficient e-voting scheme to tackle the security and privacy concerns. The proposed scheme is designed using the concept of blind signature, anonymous channel and trust worthy entities where elliptic curve cryptography (ECC) works as a backbone. The proposed scheme is formally evaluated using well known AVISPA simulation tool, which simulates the attack model using CL-AtSe as well as OFMC backend according to Dolev-Yao threat model. In both the cases, all the possible security threats are being nullified by the proposed scheme and results in safe communication. Also, the scheme is analyzed against all the possible attacks that jeopardize the integrity of the existing voting systems. Moreover, the proposed scheme is also compared with other prevailing schemes in terms of computation and communication overheads and found more efficient.

## KEYWORDS

authentication, communication security, cryptography, electronic voting and digital democracy, elliptic curve cryptography

## 1 | INTRODUCTION

Providing security to the electronic voting system is one of the prime research challenges in the modern computer science, as e-voting can play a very crucial role in a democratic setup.<sup>1-3</sup> It is a technology that will help people or electors to cast votes in an unbiased way and within a safe environment. There are various systems available for voting such as ballot voting, electronic voting machine, and so forth.<sup>1-4</sup> Conventional voting system has some disadvantages such as huge time consumption during voting process, less percentage of voting, security and integrity of the voting process, inconvenience for migrated voters, and so forth.<sup>1-6</sup> and because of that many countries like Estonia, United Kingdom and Switzerland have already adopted the e-voting system in their different referendums and elections.<sup>2-6</sup> The prime concern of e-voting system is installing trust among people about security of data and privacy of the voter which requires strong contradictory security requirements. E-voting system must comply with the following security basics like confidentiality, integrity of the voter, authentication of legitimate voter and verifiability of vote after casting.<sup>2-9</sup> These security features are essential to any e-voting system as being in a digital platform, it is vulnerable to various network attacks which may manipulate e-voting data and lead to fraud and corruption.<sup>1-3,10-13</sup> Moreover, votes of every individual voter must be counted properly to maintain integrity of the voting process and any illegal or illegitimate attempt to corrupt the process must be stopped by this system at the same time safeguarding the privacy of the voters. These are the most prominent problems and issues being faced in the present e-voting systems and not much of the research has been able to exterminate it. For many decades, the researchers are trying to find optimal solution for the security problems of e-voting and many schemes have been proposed which have taken the required measures to counter the security threats of e-voting but still some problems like authentication of legitimate user, privacy of the voters are prevailing in this context. Hence, in this paper, we have proposed a novel e-voting scheme for eradicating these above-mentioned issues of e-voting using a combination of observational and experimental techniques along with the proper security and performance analysis.

### 1.1 | Security requirements for a secure e-voting scheme

The first automated payment system was proposed in the year 1983 by Chaum et al.<sup>10</sup> Similar to payment system, e-voting concept has to ensure that electorates identity should remain secret throughout the whole voting process. Following are the basic requirements in terms of a secure electronic voting system.<sup>1-3,10,13</sup>

**Completeness:** All the valid and completed votes must be counted.

**Soundness:** The voting process should not be disrupted by adversaries.

**Fairness:** The outcome of counting must remain confidential until result declaration.

**Anonymity:** The voter's identity at any stage of the voting process must remain anonymous.

**Privacy:** In the entire voting process, individual voter's choice must remain private.

**Eligibility:** Only genuine authenticated voters should be able to vote.

**No reusability:** No elector can cast his/her vote multiple times.

**Verifiability:** The system must ensure that no adversary can tamper with a particular voter's vote and that should be verifiable.

**Efficiency:** The computation, communication, and storage overheads must be low.

### 1.2 | Related work

In this section, some of the very notable works in context of electronic voting system have been discussed. The very first ever e-voting scheme<sup>2,3</sup> based on elliptic curve cryptography (ECC) uses the trusted entities which utilize the advantages of the ECC in e-voting efficiently. However, the scheme has some weaknesses as admin or counter can destroy the votes separately.<sup>4</sup> Also, if both of them collude, the vote as well as the identity of the voter will be compromised. A voter's identity preserving secure e-voting scheme is proposed by Chen et al<sup>5</sup> where authors used certificate authority for verifying voter's identification and proxy server for preserving anonymity. This scheme<sup>5</sup> uses DLP and El Gamal encryption, which has high computation and communication overheads. Then, Khasawneh et al<sup>7</sup> have proposed a biometric based e-voting system. Another e-voting scheme using blind signature integrated with modular square root method is proposed by Yi et al<sup>8</sup> where computation complexity is low and it works in portable communication devices as well. However, the

computation overhead for encryption/decryption operation is quite high in the scheme<sup>8</sup> which is a crucial parameter in mobile e-voting system.

### 1.3 | Literature review

In this subsection, a thorough literature survey is presented where the available relevant literatures are analyzed and reviewed. In one such very notable research paper, Kouta et al<sup>9</sup> and Ibrahim et al<sup>14</sup> have used blind signature technique to tackle the previous scheme's<sup>8</sup> problem. Another scheme by Chakraborty et al<sup>6</sup> have used blind signature along with ECDLP (elliptic curve discrete logarithm problem) to enhance the security of the e-voting system. Then in 2018, Liu et al<sup>15</sup> have proposed a scheme based on secret sharing and k-anonymity that, according to the authors, ensures verifiability and protects confidentiality and anonymity as well as ensures non-cheating.

In 2019, Yi et al<sup>16</sup> have designed a block chain based scheme which uses distributed ledger technology to stop forgery as well as the authors have used ECC for authenticating and ensuring non-repudiation. The authors<sup>16</sup> have also given an option to the voter for changing his/her vote till a certain period of time. After that Li et al<sup>17</sup> have proposed a scheme based on sharing of multi-secret with multilevel access where no concept of trusted third party is involved. In the same year, another scheme by Kirilov et al<sup>18</sup> resolves the trust issues by using hyper ledger fabric. In 2020, Fan et al<sup>19</sup> have proposed an e-voting scheme based on homomorphic signcryption where number of votes is measured by signature verification. This scheme<sup>19</sup> gives privacy of voters as well as verifiability of voting. Then, Zhou et al<sup>20</sup> have presented an e-voting scheme, which uses block chain to address the limitation and weakness of the existing schemes and came up with a solution based on hyper ledger. Recently, in 2020, Khan et al<sup>21</sup> have presented a comprehensive study on performance and scalability issues of the existing e-voting schemes. Specifically, the authors<sup>21</sup> have concentrated upon different block chain settings with respect to various parameters of voting. In 2021, Boyen et al<sup>22</sup> have proposed an identity based secure e-voting scheme, which according to the authors are very efficient and secure. They have used homomorphic algorithms for encryption and decryption with that the combination of practical lattice-based cryptographic primitives makes it efficient. In, a recent work<sup>23</sup> Zhu et al have proposed a multi-district e-voting scheme based on block chain technology where authors have used two-layer block chain system where initially the voter registration will be using block chain. Then the two-layer architecture's first layer will be used to capture the voters vote of multiple districts and another layer is used for capture the vote of the electors. Woda et al in reference 24 have done a notable work in e-voting context where authors have tried to minimize the high resource requirement of block chain by using ECC and digital signature. In another work,<sup>25</sup> Mokherji et al have presented a through survey of block chain applications and how they are used in e-voting system. After that in reference 26, Nayak et al have designed a secure framework for a internet of medical things which is relevant in terms of privacy preservation in e-voting context also.

### 1.4 | Motivation and contribution

After thorough literature survey, we have observed that the previously proposed schemes are not fully protected from relevant attacks. Even if some schemes are capable of preventing attacks, they become infeasible because of high computation and communication overheads. Most of the schemes are not capable to work in the resource constraint environment that is, e-voting using a resource constraint, low power smart device which is connected to a lossy network. Schemes using block chain technology are capable for handling the security and privacy issue but due to use of block chain, a very large amount of computation and communication overhead is involved. In this context, all these above-mentioned issues have motivated us to design an efficient e-voting scheme, which is protected from any adversarial threats as well as will be lightweight. Our research contribution is as follows.

- In this paper, a novel e-voting scheme using ECC and blind signature is proposed. Both the technologies will help this scheme to be lightweight as well as gives the required level of security and voter's privacy.
- We have ensured the security strength of the proposed e-voting scheme using informal as well as formal analysis using AVISPA simulation tool.
- Furthermore, a thorough performance evaluation of the proposed scheme is done and the comparison with other relevant schemes reveals that our method provides advanced safety and privacy.

## 1.5 | Outline of the paper

The later parts of the paper are ordered as follows, Section 2 illustrates related preliminaries to understand the proposed scheme. In Section 3, the proposed method is elaborated. After that in Section 4, the security of the proposed scheme is analyzed along with results of simulation using AVISPA tool are depicted. Then in Section 5, the performance analysis of proposed scheme and comparative analysis of overheads with other related schemes are presented and finally Section 6 concludes the paper.

## 2 | PRELIMINARIES

In this section, the crucial concepts of elliptic curve cryptography (ECC) and blind signature are discussed for better understanding of the proposed scheme. Here, we will give an overview about ECC and blind signature technique.

### 2.1 | Elliptic curve cryptography

Victor Miller and Neal Koblitz have anticipated ECC in the year 1985 and 1987 respectively.<sup>11,12</sup> Later, it was accepted in public key cryptosystem by the academicians around the world. Due to the certain complexity of ECC, it has quickly mounted as most well-known and low-cost public key cryptographic arrangement which is secure, proficient in terms of both, communication as well as computation overheads.<sup>33</sup> Now, a brief definition of ECC is given where an elliptic curve  $E$  over a prime finite field  $F_p$  is established by the ensuing equation.<sup>11,12,27-29</sup>

$$Y^2 = x^3 + ax + b \quad (1)$$

Here,  $a, b \in F_p$  and the discriminant  $D = 4a^3 + 27b^2 \neq 0$ .<sup>27</sup>

The points on  $E/F_p$  defined by Equation (1) together with an extra point  $O$ , called point at infinity used for additive identity, form an additive group  $A$  as:

$$A = \{(x, y) : x, y \in F_p, E(x, y) = 0\} \cup \{O\} \quad (2)$$

Let  $n$  be the order of  $A$  which is very big and it can be defined as  $n \times G \bmod q = O$  where  $G$  is the generator of  $A$ .<sup>27</sup> Now, an additive cyclic group  $A$  will be under the point addition operation “+” defined as:  $P + O = P$ , where  $P \in A$ .

**Benefits of ECC:** ECC has following advantages, which make it more efficient and secure than its contemporary counterparts.

- ECC uses smaller key size as compared to other alike cryptosystems. For instance, RSA cryptosystem uses 1024 bits key, whereas ECC involves only 160 bits key size to guarantee same security strength.<sup>29-31</sup>
- It is nearly impossible and complex to resolve the ECC based computational problem for instance elliptic curve discrete logarithm problem (ECDLP) within polynomial time.<sup>29-31</sup> Approximately  $\sqrt{p}$  steps are essential to resolve the same where cardinality is  $p$  of that elliptic curve and which is mandatorily a large prime no.<sup>29-31</sup>
- As the key size is less in ECC based systems therefore it also has lesser and compressed message length to execute ECC-based protocols. A quicker operation consumes a smaller amount of power and therefore, ECC-based schemes are appropriate for resource-constrained entities like IoT devices.<sup>29-31</sup>

Operations that are regularly being executed in ECC. Due to the efficiency of these operations, ECC incurs low computation overhead.

- **Point addition:** Consider,  $P$  and  $Q$  be the points on Equation (1). Then,  $P + Q = R$  and  $P \neq Q$ . The line adjoining the points  $P$  and  $Q$  meets the curve in the point  $-R$  and  $R$  is the reflection of the intersecting point  $-R$ .<sup>28-31</sup>
- **Point subtraction:** Consider,  $Q = -P$ , then by replacing  $Q$  in  $P + Q = P - P = O$ . So, it is inferred that the line joining  $P$  and  $-P$  will intersect at infinity or point  $O$ .<sup>28-31</sup>

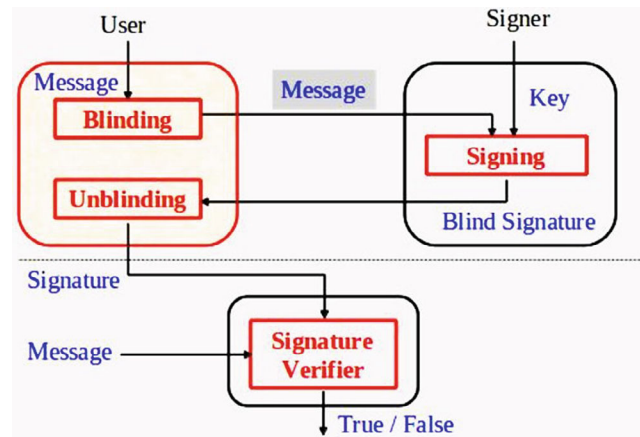


FIGURE 1 The workflow of blind signature algorithm.<sup>10</sup>

- **Point doubling:** In ECC, arithmetic point doubling is the addition of a point with itself. Here, a point  $P$  is added with itself to acquire additional point  $Q$ . Suppose,  $2P = P + P = Q$ , then according to this method, the tangent of the point  $P$  intersects the curve at  $-Q$  and the reflection of  $-Q$  in  $x$ -axis is  $Q$ .<sup>28-31</sup>
- **Scalar point multiplication:** A scalar  $t \in \mathbb{Z}_p^*$  belongs to cyclic group  $G_q$  can be multiplied with any point on the elliptic curve such that  $t \cdot P = P + P + P + \dots + P(t \text{ times}) = \sum_i^n P$ .<sup>28-33</sup>

**Computational problems of ECC:** Following computational problems of ECC are commonly used and that makes ECC very hard to compromise.

- **Elliptic curve discrete logarithm problem (ECDLP):** In cyclic additive group  $G$  if point element  $(Q)$  is given, then it is almost impossible to find the integer  $(x) \in \mathbb{Z} \times q$  such that  $Q = xP$  having  $P$  as group generator.<sup>28-31</sup>
- **Computational Diffie-Helmen problem (CDHP):** For any  $a, b \in \mathbb{Z} \times q$  it is impossible to compute  $abP$  in group  $G$  using the two given point elements  $aP$  and  $bP$  in  $G$ .<sup>28-33</sup>
- **Elliptic curve factorization problem (ECFP):** Computations of  $aP$  and  $bP$  are not feasible within polynomial time as  $P, T \in G_q$  and  $T = aP + bP$  where  $a, b \in \mathbb{Z}_q^*$ .<sup>28-33</sup>
- **Decisional Diffi-Helmen problem (DDHP):** If  $(P, aP, bP) \in G_q$  for any  $a, b, c \in \mathbb{Z}_q^*$ , it is impossible to solve  $cP = abP$  or not.<sup>28-33</sup>

## 2.2 | Blind signature

Blind signature is a special technique for digital signature where the document or message to be signed is blinded so that even the signer does not have any idea about the content of the message.<sup>10,30-35</sup> After signing, the message will be unblinded in order to check the message and verify in public with the original message for any possible distortion or tampering. This technique can be executed using various public key encryption schemes.<sup>34,35</sup> In the proposed scheme, we have shown implementation of blind signature along with the advantages of ECC. According to the blind signature scheme, it is feasible that without revealing any information about the message to the other party, a message can be get signed by the other party. This property of blind signature is used to hide voter's casted vote from the administrator and voter's identity is also not revealed ensuring the user's anonymity. Figure 1 describes the workflow of blind signature algorithm.

## 3 | PROPOSED SCHEME

In this section, a novel and competent scheme using ECC is proposed to prevent the privacy of the voter and ensure the security of the e-voting system. The proposed scheme comprises of these following entities: voter (V), certificate authority

TABLE 1 Notation used in the proposed scheme

| Notation                 | Description                                                    |
|--------------------------|----------------------------------------------------------------|
| $(V_{pub1}, V_{pri1})$   | First public/private key pair of voter (used to bind the vote) |
| $(V_{pub2}, V_{pri2})$   | Second public/private key pair of voter (used for encryption)  |
| $(A_{pub}, A_{pri})$     | Public/private key pair of administrator/authentication server |
| $(VAL_{pub}, VAL_{pri})$ | Public/private key pair of validator                           |
| $(C_{pub}, C_{pri})$     | Public/private key pair of counter                             |
| $r$                      | Random no chosen by voter                                      |
| $x$                      | Pre secret key shared by the election commission               |
| $CA$                     | Certificate authority                                          |
| $ID_V$                   | Identity of voter/user                                         |
| $B_V$                    | Biometric of the voter                                         |
| $F_q$                    | Finite field                                                   |
| $E_q(a, b)$              | An elliptic curve defined over $F_q$                           |
| $  $                     | Concatenation operation                                        |
| $?:$                     | Equal to or not                                                |
| $DC_x$                   | Digital signature issued to voter.                             |
| $DC_A$                   | Digital signature issued to administrator.                     |
| $h(.)$                   | One way hash function                                          |

(CA), authentication server /administrator, validator and counter. It is assumed that except voter, all other entities are authenticated among each other and there is no collusion among them. The proposed scheme has three phases namely (i) authentication phase, (ii) voting and validating phase and (iii) counting and acknowledgement phase. The proposed scheme uses some cryptographic tools such as digital certificates, time stamp, and blind signature. Table 1 describes the notations/symbols used in the proposed scheme along with their meaning.

Public scheme requires authenticated channels for dispersal of the public keys. In the proposed scheme, there are two aspects of authentication in e-voting system: first one is checking the user's identity in the authentication phase and second one is checking eligibility in the validation phase. A voter's identity and the right to vote are both validated and then, the casted vote is clearly separated from the identity of the voter. To be validated and authenticated, an individual has to first register as a eligible voter. Voter registration for e-voting is done in the following steps:

**Step 1:** The backbone of the proposed e-voting scheme is a client-server architecture where an application is mounted in the voter's smart device and another application tool is loaded in the administrator's server. The election commission or administrator's responsibility is to collect user's phone number, identity, address and biometric which can be finger impression or retina scan of the particular voter, and election commission will provide (voter-ID, pre-secret key) pair to the voter. These three-factors are required for authenticating a voter with administrator's server and only after that e-voting is possible through voter's mobile phone or smart device.

**Step 2:** The digital certificate will be issued after an individual sends his certificate request to CA.

**Step 3:** The digital certificate of the voter is embedded in the voter's client application (VCA).

**Step 4:** Then the voter can get ballot from organized election server where the ballot contains exclusive ids of the participating candidates.

### 3.1 | Authentication phase

In this phase, each voter is authenticated by the authentication server. At the beginning of authentication phase, an individual voter generates his/her ECC based public keys. Initially, the voter's application program picks a private key  $V_{pri1}$  where  $(1 < V_{pri1} < n \ \& \ V_{pri1} \in \mathbb{Z}_{\times q})$  and calculates its own public key  $V_{pub1}$  as  $V_{pub1} = V_{pri1} \cdot P$ . Similarly,



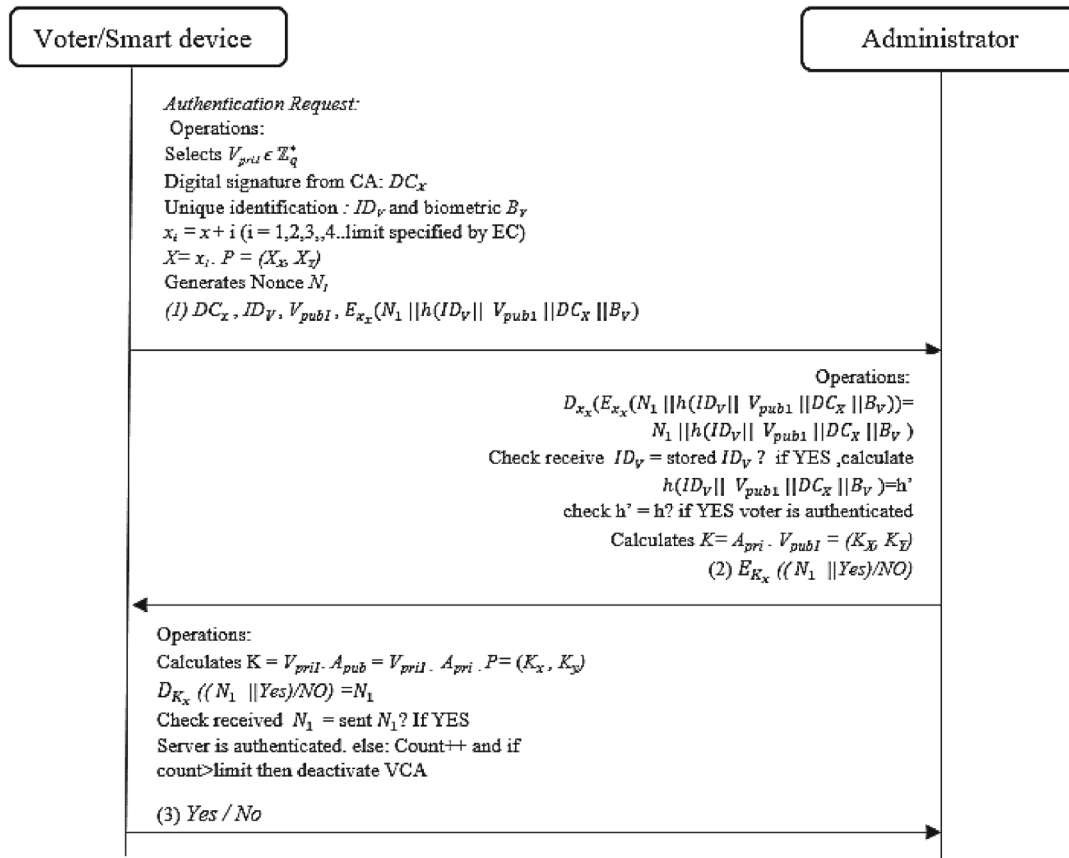


FIGURE 2 Authentication phase

authentication server also generates its public key  $A_{pub} = A_{pri} \cdot P$ . Then, the public key, hashed value of user's biometric (finger-print/retina-image) and digital signature certificate obtained from CA are transmitted after encryption to the authentication server. Finally, a session key is negotiated between the voter's application and the authentication server. The detailed procedure is depicted in step wise manner and shown in Figure 2 where  $X \rightarrow Y: Z$  means  $X$  sends message  $Z$  to  $Y$ .

Step 1: Voter  $\rightarrow$  Administrator:  $DC_x, ID_V, V_{pub1}, E_{x_x}(N_1 || h(ID_V || V_{pub1} || DC_x || B_V))$ .

The identity of voter  $ID_V$  and the digital signature certificate  $DC_x$ , public key  $V_{pub1}$ , the biometric of the voter  $B_V$  are concatenated and hashed. After that nonce  $N_1$  (a test for authentication between user and server and to negotiate session key) is generated and concatenated with the previous hash. Then this whole message is encrypted by  $x$ -coordinate of  $X$ , where  $X = x_i$ ,  $P$  and  $x_i = x + i$  such that  $i = 1, 2, \dots, n$ . As the session key is generated with the pre shared secret key by the election commission, the direct use does not compromise the pre-shared secret and that's why it is periodically updated and incremented where the no of increment is determined by the election commission. Finally, the encrypted message along with  $DC_x, ID_V, V_{pub1}$  is sent to the authentication server.

Step 2: Administrator  $\rightarrow$  Voter:  $E_{K_x}((N_1 || Yes)/NO)$

After receiving the message, the administrator decrypts the communication and gets the challenge  $N_1$  and  $h(ID_V || V_{pub1} || DC_x || B_V) = h$ . First, it verifies the  $ID$  information from the stored information. If  $ID$  is valid, the server calculates hash digest with stored information and checks whether  $h' = h$ . If it is unsuccessful, the authentication process is concluded and a message with  $NO$  is sent to the voter and if it succeeds, the user application program generates the session key  $K = A_{pri} \cdot V_{pub1} = (K_x, K_y)$ . A message with nonce  $N_1$  is sent with a  $YES$  message, encrypted with the generated session key, and sent to the voter's application.

Step 3: Voter  $\rightarrow$  Administrator:  $YES/NO$

Voter's application decrypts the message by generating the same session key  $K = V_{priI} \cdot A_{pub} = V_{priI} \cdot A_{pri} \cdot P = (K_x, K_y)$ . Then checks the nonce value with the initial nonce value. If it is found correct, the server is also authenticated else update the pre secret key and reinitiate the process until it reaches the maximum attempts. If it receives a  $NO$ , the session is immediately terminated.

### 3.2 | Voting and validating phase

In this phase, we have described the secure voting process. This phase deals with these following issues: (i) only eligible voter is allowed to vote (ii) multiple voting is restricted and this phase runs parallel with the voting phase that is, once the voting is done, it is first validated using the steps of this phase and then it will move forward with the rest of the operations of the voting phase. To successfully accomplish this phase, some steps need to be followed that are discussed below in step-by-step method where  $A \rightarrow B: C$  means A sends communication C to B. The summary of this phase is shown in Figure 3.

Step 1: Voter  $\rightarrow$  Admin:  $E_{K_x}(E_{V_{pri2}}(E_{V_{pub1}}(m \times r) || ID_V || DC_X))$

Now, the voter's application selects a random number  $r$  and candidate  $ID$  from the list of the candidates according to their constituencies and whom the voter wants to vote. The digital certificate of the candidate is embedded with the  $ID$ . First, the VCA (voter client application) will verify the digital certificate. After that in order to prevent the admin to see the vote, VCA first blinds the vote  $m$  as  $E_{V_{pub1}}(m \times r)$ . Then, it concatenates its id  $ID_V$  and digital certificate  $DC_X$  and further encrypts it with voter's second private key  $E_{V_{pri2}}$  and finally with negotiated shred session key  $K$ , so that only admin can decrypt the message and verify voter's identity.

After getting the message from VCA, the administrator decrypts the message using its shared secret key  $K$  and again decrypts using voter's second public key and obtain  $E_{V_{pub1}}(m \times r) || ID_V || DC_X$  containing the voter's id, digital certificate, and blinded vote. Admin first verifies the  $ID$  with the calculated ID. If validated then generates *pseudoID*.

Step 2: Admin  $\rightarrow$  Validator:  $E_{K_{x1}}(E_{A_{pri}}(ID_V) || DC_A || E_{V_{pub1}}(m \times r))$

Now, admin generates a shared session key with the validator  $K = A_{pri} \cdot VAL_{pub} = A_{pri} \cdot VAL_{pri} \cdot P = (K_{x1}, K_{y1})$  and requests validator for validating the eligibility of the voter by encrypting the voter's id concatenated with its own digital certificate and the encrypted casted vote and further encrypts with the shared key and sends the message to the validator.

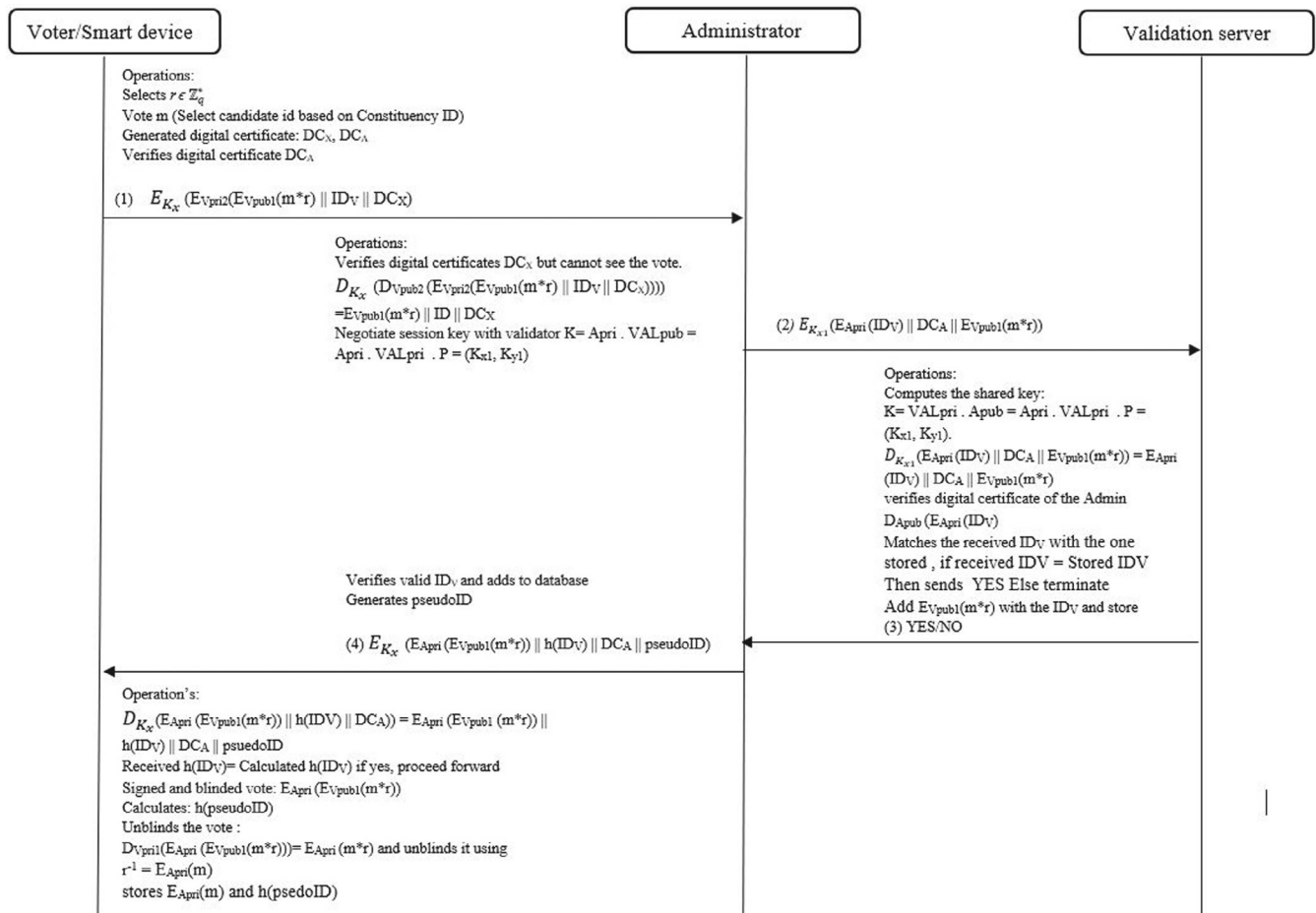


FIGURE 3 Summary of voting and validation phase



Step 3: Validator  $\rightarrow$  Admin: YES/NO

Upon receiving the message, validator decrypts the message with the generated shared key resulting in  $E_{Apri}(ID_V) || DC_A || E_{Vpub1}(m \times r)$ . Then the validator first decrypts the identification details, checks with the validator's database for eligibility and if found eligible, adds the encrypted part of the vote  $E_{Vpub1}(m \times r)$  to the corresponding  $ID$  which ensures that double voting does not take place. If validated and have not voted before then the validator sends a message YES to the admin otherwise NO will be sent and session is terminated and the vote will not be registered.

Step 4: Admin  $\rightarrow$  Voter:  $E_{K_x}(E_{Vpub1}(E_{Apri}(m \times r)) || h(ID_V) || DC_A || pseudoID)$

Upon receiving the validation confirmation from validator, the admin encrypts the blinded vote using server's private key, concatenates the hashed  $ID$ ,  $h(ID_V)$ , the digital certificate of admin and the  $pseudoID$ . Then, encrypts with voter's second public key and finally encrypts with shared key  $K$  in order to make sure only voter can decrypt. Then, admin sends this message to VCA.

On receiving the message, VCA decrypts the message using its shared key  $K$  and obtains  $(E_{Vpub1}(E_{Apri}(m \times r)) || h(ID_V) || DC_A || pseudoID)$ . Then, validates the  $ID$  with the received value. If validated, then proceeds further otherwise terminates the session. Now, the VCA calculates the hash of  $pseudoID$  as  $h(pseudoID)$  and then unblinds the vote using its private key and multiplying with  $r^{-1}$  which results in two final message (i) signed vote receipt from admin  $E_{Apri}(m)$  and (ii)  $h(pseudoID)$  which VCA will store.

### 3.3 | Counting and acknowledgement phase

In this phase, counting of the vote and providing acknowledgement to the voter is executed. In this phase, we are considering the following points: (i) every vote is counted by the counter, (ii) an acknowledgement is needed to be sent to the voter about its voting information, except his casted vote, along with the time stamp at which he voted for transparency, (iii) counter should not know the voter's identity but only his/her vote such that counter and admin both together are unable to harm the system. To successfully accomplish these tasks, the required process is designated here in stepwise manner where  $P \rightarrow Q:R$  means  $P$  directs message  $R$  to  $Q$ . The summary of this phase is depicted in Figure 4.

Step 1: Voter/smart device  $\rightarrow$  Counter:  $E_{K_{x2}}(E_{Cpub}(E_{Apri}(m) || h(pseudoID)))$

Immediately after the user casts his/her vote in the voting phase, the stored  $E_{Apri}(m)$  and  $h(pseudoID)$  is concatenated and sent to the counter for counting after encryption using the public key of the sender. ECC based shared session key between VCA and counter is calculated by the voter's smart device as  $K = V_{pri1} \cdot C_{pub} = V_{pri1} \cdot C_{pri}$ .  $P = (K_{x2}, K_{y2})$ . Finally, the message  $E_{K_{x2}}(E_{Cpub}(E_{Apri}(m) || h(pseudoID)))$  is sent to the counter.

Upon receiving the message, the counter calculates its secret session key  $K = C_{pri} \cdot V_{pub} = C_{pri} \cdot V_{pri1}$ .  $P = (K_{x2}, K_{y2})$  and decrypts the received message  $D_{K_{x2}}(E_{K_{x2}}(E_{Cpub}(E_{Apri}(m) || h(pseudoID))))$ . The counter further decrypts the communication with its own private key  $C_{pri}$  and administrator's public key  $A_{pub}$  and gets the signed vote and hash digest of  $pseudoID$ , which will be stored as  $h1$ . The counter also verifies the admin's signature by decrypting it using admin's public key resulting in the vote  $m$ , which it stores for counting.

Step 2: Admin  $\rightarrow$  Counter:  $E_{Cpub}(h(pseudoID))$

Simultaneously, the admin sends the hash of the  $pseudoID$ , which it had generated after casting of the vote and it is being encrypted using counter's public key and sent to the counter.

Upon receiving the message, the counter decrypts it using its private key and stores the hash digest as  $h2$ . Then the counter checks whether  $h1 = h2$ . If it is true then it means that voter's vote is counted and counter records the timestamp.

Step 3: Counter  $\rightarrow$  Voter/ smart device:  $E_{K_{x2}}(Ack, T_1, pseudoID)$

Counter sends an acknowledgment to the voter along with the  $pseudoID$  and timestamp by encrypting it using the shared key. Another acknowledgement is sent to the admin as well, else the whole process is canceled, and the vote will not be counted.

Upon receiving the message, the voter's application decrypts it using the same shared key as  $D_{K_{x2}}(E_{K_{x2}}(Ack, T_1, PseudoID))$  and stores the information as proof of successful voting.

## 4 | SECURITY ANALYSIS

This section, presents the complete security analysis of our proposed scheme. Informal security analysis using mathematical and logical assumptions as well as formal security analysis using AVISPA simulation tool. This security analysis proves that our scheme is safe from the relevant security threats.

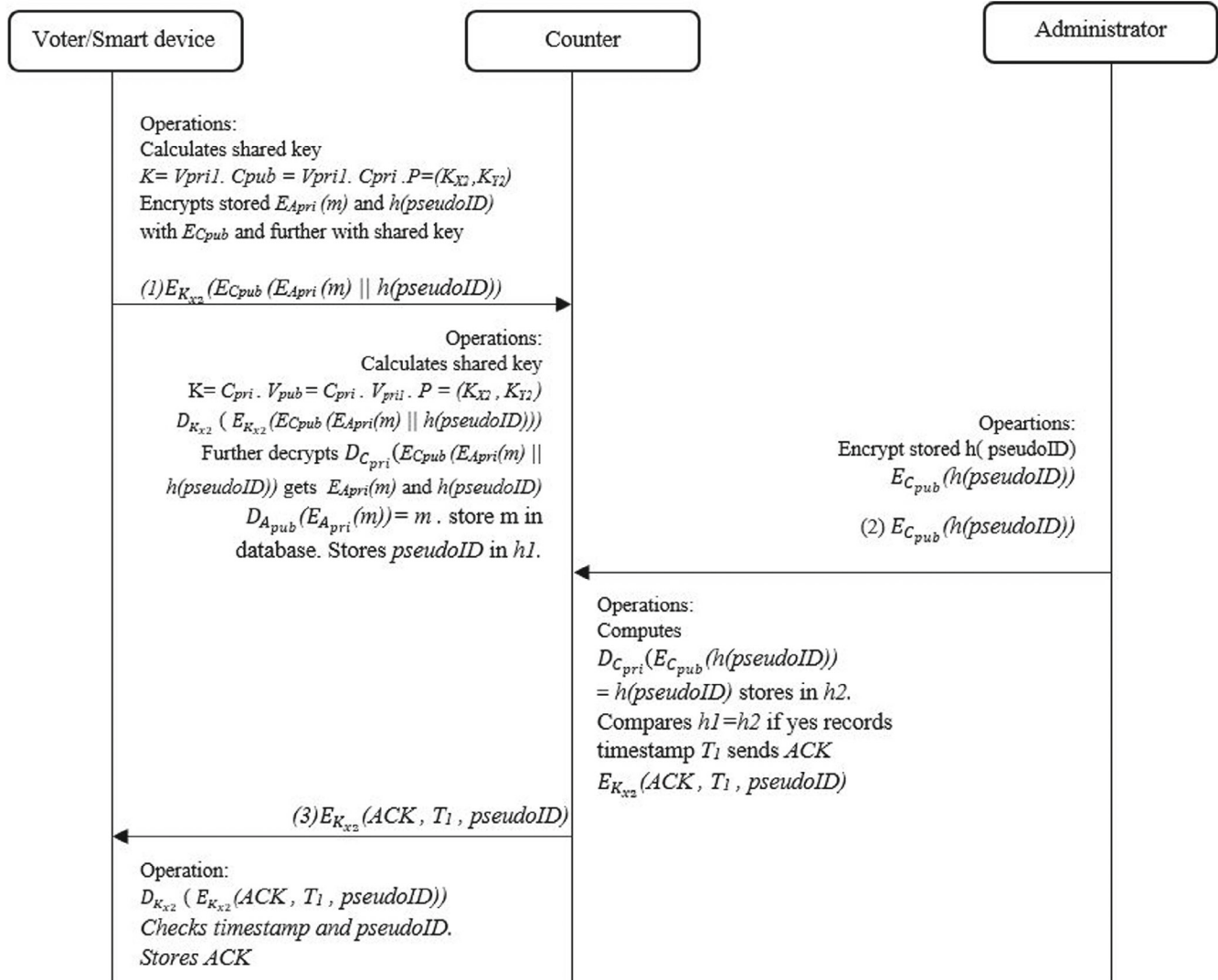


FIGURE 4 Counting and acknowledgement phase

#### 4.1 | Informal security analysis

Informal analysis of the proposed scheme is being discussed in this subsection. To prove that the proposed scheme is safe from all the relevant security attacks and achieves the desired security goals, we have used mathematical and logical reasoning.

**Proposition 1.** *Proposed system is safe from man-in-the-middle attack.*

*Proof.* In the proposed e-voting scheme, a voter already has his/her integrated digital signature from the CA that is further used to help other parties to verify the public keys before message encryption. Therefore, it has been sent along with its given identification ID (such as Aadhaar ID is used as a supplement for biometric details) and public key of the user as a three-factor authentication operation. The primary idea of man-in-the-middle attack is that the adversary pretends to be a user and creates two sessions that is, with the voter and server separately for misleading them and intercepting the message but since the keys of all the parties are being authenticated, it further makes the scheme less susceptible to this attack.

Message for user authentication  $E_{Xx}(N_1 || h(ID_V || V_{pub1} || DC_X))$  which comprises user-ID, user's public key and user's digital signature, consequently none of these can be counterfeit and altered.

Message received by server on decryption of  $D_{Xx}(E_{Xx}(N_1 || h(ID_V || V_{pub1} || DC_X)))$  has challenge  $N1$  and hash digest of user-ID, user public key and user digital signature. This is tested against its self-calculated hash digest for authentication

and if the attacker tries to modify message  $E_{xx}(N_1 || h(ID_V || V_{pub1} || DC_X))$  using MIM attack, it can be easily detected by the administrator. ■

**Proposition 2.** *Proposed scheme is safe from replay attack.*

*Proof.* In a replay attack, an adversary who might have eavesdrop on the channel and collected some information about previous session, reinitiates the communication between entities. The proposed scheme is protected from this attack as the message for authentication is encrypted by a value  $x + i$  (where  $i$  is incremented with every unsuccessful attempt) and whose value cannot be guessed by the adversary due to the use of ECC. So, adversary might try to modify it by sending message  $(x + j)$  (where  $j \neq i$ ). Hence, when this message is decrypted, it will be found incorrect since the digital signature and hash value are different and so server will terminate this authentication and login attempt. Therefore, the proposed scheme is resilient to replay attack. ■

**Proposition 3.** *The proposed scheme is resistant to DoS attack.*

*Proof.* Since a limited number of attempts are available for a user to cast his/her vote and if user fails to get itself authenticated within these many numbers of attempts, his/her certificate will be deactivated for certain period of time and will be revoked later. Therefore, an attacker will never be able to send request more than this limited number, hence resisting the DoS attack. ■

**Proposition 4.** *The proposed scheme is secure from impersonation attack.*

*Proof.* The impersonation attack is possible in the following two ways: (1) user spoofing and (2) server spoofing and none of these are achievable in the proposed scheme.

Let us assume that an attacker wants to impersonate a voter's application by sending its own public key  $V_{pub}$ , knows a valid  $ID_V$ . He/she would create his/her own message digest and send to the server but the server while rechecking the hash digest will find a mismatch and will be able to identify that the user is not an authenticated user and will terminate the process instantly. Then on the other hand, the authenticity of the public key is being verified by the digital certificates issued to the participating entities, the spoofing of the server is quite a difficult process since the server is encrypting the messages using the voter's original certified public key. So, no one except the voter can modify or intercept the nonce  $N_1$  and hence verifying the server's identity successfully, server spoofing is prevented. ■

**Proposition 5.** *Proposed scheme can maintain perfect forward secrecy.*

*Proof.* The perfect forward secrecy means the compromise of the long-term private keys of both the participating entities should not affect the security of the previously shared short-term pre secret key. In the proposed scheme, the client is encrypting the data using the pre secret key shared during registration and disclosure of the private keys will only help the attacker to know the nonce value  $N_1$ , which will be of no further use to the attacker. Hence, it maintains perfect forward secrecy. ■

**Proposition 6.** *Proposed scheme supports non-repudiation.*

*Proof.* Since the public keys of the communicating entities is being first certified and communications are done using digitally signed certificates so that the participating entities are not able to deny the sending of a message. However, these security measure can fail in some instance therefore we have also captured identification details of the voter (with the integrated biometric details along with such as Aadhaar-ID) that would collectively become difficult to repudiate. ■

## 4.2 | Formal security analysis

In this section, the formal security analysis of the proposed scheme is presented. Here to formally analyze the proposed scheme, we have used AVISPA. AVISPA is a regularly used testing toolkit used for the evaluation of certain safety protocol specifications.<sup>36</sup> AVISPA uses a language for the role-oriented definition of protocol. The AVISPA is an abbreviation, which means automated validation of Internet security protocols and applications. It provides a segmental and communicative prescribed language for describing the protocols and their security characteristics through push-button tool.

Also it has different back-ends that participate in a variety of automated scheme analysis tools.<sup>29,37,38</sup> The HLPSP is the abbreviated form of high-level protocol specification language. HLPSP is used to creating security features by designing the security protocols accordingly the scheme that is proposed. The specified code for a particular security scheme in HLPSP is translated into the intermediate code using a translator termed HLPSP2IF.<sup>29,37,38</sup> This intermediate code is read straight by the AVISPA tool's back-ends and which uses a separate language than HLPSP. Later on, the description mentioned in intermediate code is used as input to the back-ends of AVISPA tools. As a result, the tools can analyze according to the security requirements and conditions.<sup>29,37,38</sup> AVISPA back-ends consists of four tools. Those are OFMC (On the fly model checker), CL-AtSe (CL-based Attack Searcher), SATMC (SAT-based model checker), and TA4SP (Tree automata-based protocol analyzer).<sup>29,37,38</sup> The HLPSP codes for different roles used to simulate the proposed protocol are given in Appendix A as Figures A1–A6.

#### 4.2.1 | Simulation results

The simulation results for the proposed scheme using AVISPA are shown in Figures A7 and A8 given in the Appendix A, which clearly proves that the proposed e-voting scheme is secure from all the pertinent security threats. The scheme is specified under two back-ends of AVISPA using Security Protocol Animator (SPAN)<sup>30–32</sup> namely OFMC and CL-AtSe which are broadly used in various security schemes. As mentioned above in Section 4.2, the specified code in HLPSP is converted into IF to generate the final report. The Output Format (OF) of the generated report is structured as stated below.

- **Summary:** It shows the final outcome of the analysis; whether it is —SAFE or UNSAFE
- **Details:** This presents the detailed information about the stipulation, determining if any attacks can happen, what models are employed and finally, why the analysis was inconclusive.
- **Protocol, Goal, Backend:** It depicts the name of the protocol, specified goals, and back-end respectively.
- **Statistics:** It shows the time taken for executing the protocol by the specified back-end.
- **Attack trace:** It depicts how (if any) attack is carried out on the protocol.

The obtained simulation results exhibited in Figures A7 and A8, which confirm that safety of the protocol against the two aforesaid attacks. OFMC and CL-AtSe back-ends perform the following verifications of the scheme:

- **Implement-ability verification on significant HLPSP specifications:** The executing ability of the protocols is most essential since it depicts that whether the protocol is implementable or not. It may happen that the specified protocol is not properly modeled so that the AVISPA could not find any result from the protocol specification. In the Figures A7 and A8, it is clearly visible that our scheme accomplishes all the required goals as specified and passes the implement ability check successfully.
- **Replay attack test:** The OFMC and CL-AtSe, the both backend validate whether the genuine agents are able to execute the specified protocol by achieving a search of a passive intruder. This backend offer the normal knowledge to the intruder of any common session between the valid agents. The verification result given in Figures A7 and A8 undoubtedly indicates that our scheme is secure against the replay attack.
- **Dolev-Yao model verification:** The OFMC backend of AVISPA confirms if there exists possibility of occurrence of any man-in-the-middle attack by an active intruder. The result is visibly shown in the In this section simulation results of HLPSP code have been represented in Figures A7 and A8 which depict the results for the back-ends OFMC and CL-AtSe of AVISPA. The simulation results authorize that the proposed scheme is SAFE and able to withstand all active and passive attacks like man-in-the-middle attack, replay attacks, and so forth.

In this section simulation, result of HLPSP code has been represented in Figure A7, which depicts the results for the back-ends OFMC and CL-AtSe of AVISPA. The simulation results authorize that the proposed scheme is SAFE and able to withstand all active and passive attacks like man-in-the-middle attack, replay attacks, and so forth.

## 5 | PERFORMANCE ANALYSIS

This section presents and evaluates the complete efficiency of the proposed scheme. The performance assessment is carried out through comparison of computation and communication overheads as well as comparing the security features of the proposed scheme with other relevant security structures in this context.

### 5.1 | Comparison of computation overheads

In this subsection, the computation overhead of the proposed scheme is being compared with other relevant schemes. While calculating the computation overheads, we have considered the authentication and voting phase only and used the same experimental setup used in Ahmad et al,<sup>2</sup> Sadhukhan et al,<sup>30</sup> Rangwani et al,<sup>31</sup> Chatterjee et al<sup>29</sup> and Sowjanya and Majumdar et al in references 37,38 and 39 Sahzad et al respectively. The time required for different cryptographic operations with their symbols are given in Table 2. In this work, we have compared our proposed scheme with<sup>40-42</sup> schemes which are very relevant and recent literature in this context. Table 3 depicts the said comparison of computation overhead calculation. After thorough observation of the schemes<sup>40-42</sup> we have calculated the total time needed by summing up the individual time required for different cryptographic operations mentioned in Table 2. Our proposed scheme uses two one-way hash operations, two ECC based encryption and decryptions and one blind signature operation. Hence, the total time required is 11.0246 ms, which is clearly the least among the other relevant schemes. As we have used ECC based encryption/decryption and lightweight hash function, the overall computation overhead is less than other relevant schemes. It is well recognized that the requisite time for encryption/decryption operation for symmetric ECC-160 bit is relatively lesser than that of RSA-1024 bit asymmetric key encryption/decryption. Also, due to elliptic curve logarithmic computational hardness it is very difficult for an adversary to break our ECC based scheme. The comparison of computation overheads is shown in Table 3. The comparison between our proposed scheme and other relevant schemes in terms of security features are given in the Table 4. As it can be seen that schemes in references 40-42 does not meet all the security requirements necessary for e-voting hence it can be said that our proposed scheme is superior in terms of efficiency and security strength than other relevant schemes.

**TABLE 2** Symbols for cryptographic operations with explanation

| Symbols   | Explanation                                    | Time (in ms) |
|-----------|------------------------------------------------|--------------|
| $T_{ECC}$ | Time for ECC-160 Encryption/ Decryption        | 3.85         |
| $T_{SM}$  | Time required for point multiplication         | 2.226        |
| $T_{PA}$  | Time required for point addition               | 0.0288       |
| $T_{ME}$  | Time for modular exponentiation(1024 bits)     | 3.85         |
| $T_H$     | One-way hash function (SHA-1256)               | 0.0023       |
| $T_{E/D}$ | Time requirement for RSA-1024 encryption       | 3.85         |
| $T_C$     | Time requirement for digital comment operation | 0.83         |
| $T_B$     | Time required for a blind signature operation  | 3.32         |
| $T_S$     | Time required for RSA signature operation      | 3.85         |

**TABLE 3** Comparison of computation overhead

| Scheme                        | Voter                                 | Total   |
|-------------------------------|---------------------------------------|---------|
| Ye et al <sup>40</sup>        | $21T_{ME} + 5T_{SM} + 7T_{PA} + 4T_H$ | 92.16   |
| Darwish's et al <sup>41</sup> | $2T_{ME} + 6T_S + 1T_B + 1T_D$        | 37.97   |
| Zhou et al <sup>42</sup>      | $1T_{ME} + 2T_S + 1T_B + 1T_C$        | 18.72   |
| Proposed scheme               | $1T_H + 2T_{ECC} + 1T_B$              | 11.0223 |



**TABLE 4** Comparison of security features of different security schemes

| Security features   | Proposed scheme | Ye et al <sup>40</sup> | Darwish's et al <sup>41</sup> | Zhou et al <sup>42</sup> |
|---------------------|-----------------|------------------------|-------------------------------|--------------------------|
| Completeness        | YES             | YES                    | YES                           | YES                      |
| Uniqueness          | YES             | YES                    | YES                           | YES                      |
| Fairness            | YES             | YES                    | YES                           | YES                      |
| Anonymity           | YES             | YES                    | YES                           | YES                      |
| Privacy             | YES             | YES                    | YES                           | NO                       |
| Eligibility         | YES             | YES                    | YES                           | YES                      |
| Coercion resistance | YES             | YES                    | NO                            | YES                      |
| Verifiability       | YES             | YES                    | YES                           | YES                      |
| Efficiency          | HIGH            | LOW                    | LOW                           | HIGH                     |

From the above comparison, it is clear that the proposed scheme has least computation overhead than the other relevant schemes.

## 5.2 | Comparison of communication overheads

This subsection discusses the communication overhead of the different relevant schemes and compares them with the proposed scheme. To analyze communication overhead, we will look at the number of messages exchanged between different entities together with the size of the messages. Here, we have considered the communication overhead of authentication phase only. The proposed scheme makes use of 160 bit key for ECC and because of that the message size is smaller than most of the other schemes which use 1024 bits key for RSA. In our scheme, the authentication phase has only 2 messages exchanged that is much less than all other relevant schemes.<sup>36-39</sup> So, we can say that the proposed scheme is more efficient in terms of communication overhead.

Finally, the comparison of security features with other relevant schemes confirms the efficiency of the proposed scheme and shown in Table 4.

## 6 | CONCLUSION AND FUTURE WORK

A novel ECC based e-voting scheme is introduced in this paper that satisfies almost all the security requirements. With this electronic voting scheme, the voting process will be much faster and safer than the traditional voting system. Less chances of tampering of vote / foul play are there in this method of voting. The only concern in e-voting system as a whole is the security of scheme so that no adversary can breach the privacy of the voter. In this context, to make this process secure and keep the user's privacy protected, the proposed scheme is proved to be very efficient as it uses ECC based approach and blind signature algorithm. Using informal security analysis and AVISPA simulation, it has been shown that the proposed scheme is safe from all the relevant security attacks. Then by comparing computation and communication overheads, it is proven that the effectiveness of the proposed scheme is also very high. Moreover, the security features comparison clearly establishes that the proposed scheme is the best suitable solution in the e-voting context. As the future work of the proposed scheme, an additional reduction of computation and communication overhead can be considered using certificate less cryptography or using identity-based encryption techniques. Use of Block chain and machine learning techniques in conjunction with our proposed authentication and key management approach can raise the security strength of the overall communication. So, the design of such schemes, implementing and evaluating them in the real world scenario can be considered as a potential future work towards making an efficient and secure e-voting system.

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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## APPENDIX A. The HPSL codes of AVISPA simulation for different roles

```

role voter(V,A,VS,C: agent,
           X,K,Epub: symmetric_key,
           H: hash,
           Mul,Add: function,
           SND,RCV: channel(dy))
played_by V
def=
local State : nat,
VPU,VH,VB,IDV,IDB,P,S,RH,T1,T2,M:
text,
C1,C2,C3,C4,RHH,RBB,MH,SK:message,
Inc : hash
const
auth_1,auth_2,auth_3,auth_4,auth_5,a
uth_6,
subs1,subs2,subs3,subs4,subs5,subs6,
subs7,subs8:protocol_id
init State := 0
transition
1. State=0 /\ RCV (start) =|>
   State' :=1 /\ UH' := new()
   /\ VH' := Mul (UH'.P)
   /\ X' := Mul (UH'.VB)
   /\ C2' :=Mul (UH'.P)
   /\ SND(C1'.RHH'.C2')
   /\ secret ({UH'},subs1,{A})
   /\ secret ({S'},subs2,{A})
   /\ secret ({RH'},subs3,{A})
   /\ secret ({RHH'},subs4,{A,B})
   /\ request (A,B, auth_1, C2)
2. State = 1 /\ RCV (RBB'.C3')=|>
   State' := 2 /\ C3'
:=H(IDB.C1'.RBB.RHH'.K') % line 32
   /\ witness(A,B,auth_2,C3')
   /\ SK' := H(Mul(RH'.RBB))
   /\ SND({C3.M}_SK')
   /\ request(A,B,auth_3,C3)
   /\ secret ({SK},subs5,{A,B})
end role

```

**FIGURE A1** HPSL code for role voter

```

role Administrator( A,V,VS,C: agent,
    X,K,Epub: symmetric_key,
    H: hash,
    Mul,Add: function,
    SND,RCV: channel(dy))
played_by A
def=
local State : nat,
UB,VH,VB,IDH,IDB,P,S,RB,T1,T2,M:
text,
C1,C2,C3,C4,RHH,RBB,MH,SK:message,
Inc : hash
const
auth_1,auth_2,auth_3,auth_4,auth_5,auth_6,
subs1,subs2,sub3,sub4,sub5,subs6,subs7,subs8:protocol_id
init State := 0
transition
1. State=0 /\ RCV (C1'.RHH'.C2') =|>
    State' :=1 /\ UB' := new()
    /\ VB' := Mul (UB'.P)
    /\ K' := Mul (UB'.VH)
    /\ MH' :=H (C1'.RHH'.IDB.K')
    /\ SND (C3'.RBB')
    /\ secret ({UB'},subs6,{B})
    /\ secret ({RB'},subs7,{B})
    /\ secret ({RBB'},subs8,{A,B})
    /\ request (B,A,auth_5, C3')
%line 67
2. State = 1 /\ RCV ({C4.M}_SK') =|>
    State' :=2 /\ SK' :=
H (Mul (RB'.RHH'))
    /\ witness (B,A, auth_6, C3')
end role

```

FIGURE A2 HLPSL code for role administrator

```

role session((C,VS,A,V: agent,
    X,K,Epub: symmetric_key,
    H: hash,
    Mul,Add: function,
    SND,RCV: channel(dy))
def=
local SND1,SND2,RCV1,RCV2: channel
(dy)
composition
voter
(V,A,VS,C,X,K,Epub,H,Mul,Add,SND1,RCV2)/\adminster(V,A,VS,C,X,K,Epub,H,Mul,Add,SND1,RCV2)/\votingserver(V,A,VS,C,X,K,Epub,H,Mul,Add,SND1,RCV2)/\counter(V,A,VS,C,X,K,Epub,H,Mul,Add,SND1,RCV2)
end role

```

FIGURE A3 HLPSL code for role session



```

role Votingserver( VS,A,V,C: agent,
  X,K,Epub: symmetric_key,
  H: hash,
  Mul,Add: function,
  SND,RCV: channel(dy))
played_by VS
def=
local State : nat,
UB,VH,VB,IDH,IDB,P,S,RB,T1,T2,M:
text,
C1,C2,C3,C4,RHH,RBB,MH,SK:message,
Inc : hash
const
auth_1,auth_2,auth_3,auth_4,auth_5,a
uth_6,
subs1,subs2,sub3,sub4,sub5,subs6,sub
s7,subs8:protocol_id
init State := 0
transition
1. State=0 /\ RCV (C1'.RHH'.C2') =|>
  State' :=1 /\ UB' := new()
  /\ VB' := Mul (UB'.P)
  /\ K' := Mul(UB'.VH)
  /\ MH' :=H(C1'.RHH'.IDB.K')
  /\ SND(C3'.RBB')
  /\ secret ({UB'},subs6,{B})
  /\ secret ({RB'},subs7,{B})
  /\ secret ({RBB'},subs8,{A,B})
  /\ request (B,A,auth_5, C3')
%line 67
2. State = 1 /\ RCV ({C4.M}_SK') =|>
  State' :=2 /\ SK' :=
H(Mul(RB'.RHH'))
  /\ witness(B,A, auth_6, C3')
end role

```

FIGURE A4 HLPST code for role validator

```

role Counter(C,VS,A,V: agent,
  X,K,Epub: symmetric_key,
  H: hash,
  Mul,Add: function,
  SND,RCV: channel(dy))
played_by VS
def=
local State : nat,
UB,VH,VB,IDH,IDB,P,S,RB,T1,T2,M:
text,
C1,C2,C3,C4,RHH,RBB,MH,SK:message,
Inc : hash
const
auth_1,auth_2,auth_3,auth_4,auth_5,
auth_6,
subs1,subs2,sub3,sub4,sub5,subs6,su
bs7,subs8:protocol_id
init State := 0
transition
1. State=0 /\ RCV (C1'.RHH'.C2')
=|>
  State' :=1 /\ UB' := new()
  /\ VB' := Mul (UB'.P)
  /\ K' := Mul(UB'.VH)
  /\ MH' :=H(C1'.RHH'.IDB.K')
  /\ SND(C3'.RBB')
  /\ secret ({UB'},subs6,{B})
  /\ secret ({RB'},subs7,{B})
  /\ secret ({RBB'},subs8,{A,B})
  /\ request (B,A,auth_5, C3')
%line 67
2. State = 1 /\ RCV ({C4.M}_SK')
=|>
  State' :=2 /\ SK' :=
H(Mul(RB'.RHH'))
  /\ witness(B,A, auth_6, C3')
end role

```

FIGURE A5 HLPST code for role counter

```

role environment ()
def=
const v,a,vs,c: agent,
k,x,epub: symmetric_key,
h: hash,
mul, add: function,
uh,vh,vb,idh,idb,p,s,rh,t1,t2,ub,rb,m: text,
auth_1,auth_2,auth_3,auth_4,auth_5,auth_6,
subs1,subs2,sub3,sub4,sub5,subs6,subs7,subs8:protocol_id
intruder_knowledge={a,b,idh,idb,h,mul,add}
composition
session (a,b,k,h,mul,add) /\ session (b,a,k,h,mul,add)
end role
goal
secrecy_of subs1
secrecy_of subs2
secrecy_of subs3
secrecy_of subs4
secrecy_of subs5
secrecy_of subs6
secrecy_of subs7
secrecy_of subs8
authentication_on auth_1_C2
authentication_on auth_2_K
authentication_on auth_3_C3
authentication_on auth_4_C3
authentication_on auth_5_C3
authentication_on auth_6_C3
end goal
environment()

```

**FIGURE A6** HLPSL code for role environment

The output of the AVISPA simulation for CL-ATSE and OFMC background

```

SUMMARY
SAFE

DETAILS
BOUNDED_NUMBER_OF_SESSIONS
TYPED_MODEL

PROTOCOL
/home/span/span/testsuite/results/proto1.if

GOAL
As Specified

BACKEND
CL-AtSe

STATISTICS

Analysed : 0 states
Reachable : 0 states
Translation: 0.00 seconds
Computation: 0.00 seconds

```

**FIGURE A7** CL-ATSE backend result of AVISPA

```
% OFMC
% Version of 2006/02/13
SUMMARY
  SAFE
DETAILS
  BOUNDED_NUMBER_OF_SESSIONS
PROTOCOL
  /home/span/span/testsuite/results/proto1.if
GOAL
  as_specified
BACKEND
  OFMC
COMMENTS
STATISTICS
  parseTime: 0.00s
  searchTime: 0.00s
  visitedNodes: 5 nodes
  depth: 2 plies
```

**FIGURE A8** OFMC backend result of AVISPA